Exam 01

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1 Exam Metadata

2 Part A: MULTIPLE CHOICE QUESTIONS

- **Instructions**: There is only one correct option for each question. Each problem is worth 1 point for a completely correct answer. No partial credit is given.

2.1 Problem A1

The equation $z^2 + iz - 1 = 0$ has two roots z_1 and z_2. Find $z_1^2 + z_2^2$.

- ☐ (a) -1
- ☐ (b) 0
- ☐ (c) 1
- ☐ (d) 4
- ☐ (e) 12

2.2 Problem A2

What will the following code output? The command `a % b` denotes the remainder from dividing `a` by `b`.

```
s = 0
for i in 1:11
    if i % 3 == 0
       s += i
    elseif i % 5 == 0
       s -= i
    end
end
println(s)
```

- ☐ (a) -2
- ☐ (b) 3
- ☐ (c) 4
- ☐ (d) 13

2.3 Problem A3

Let $A = \{k \in \mathbb{N} : k < 20\}$, $B = \{3k - 1 : k \in \mathbb{N}\}$, and $C = \{2k + 1 : k \in \mathbb{N}\}$. Determine the cardinality of the set $(A \cap B) \setminus C$.

- ☐ (a) 2
- ☐ (b) 3
- ☐ (c) 4
- ☐ (d) 8

2.4 Problem A4

The set A has 10 elements, the set B has 14 elements. Choose the correct statement.

- ☐ (a) The number of injective functions from A to B is greater (strictly) than the number of surjective functions from A to B.
- ☐ (b) The number of surjective functions from A to B is greater (strictly) than the number of bijective functions from A to B.
- ☐ (c) The number of injective functions from A to B is equal to the number of injective functions from B to A.

2.5 Problem A5

Given propositions P and Q, when the proposition $((P \implies Q) \wedge Q) \implies P$ is false?

- ☐ (a) When P is false and Q is true.
- ☐ (b) When P is false and Q is false.
- ☐ (c) When P is true and Q is false.
- ☐ (d) When P is true and Q is true.

2.6 Problem A6

Choose a pair of orthogonal planes in $\mathbb{R}^3$:

- ☐ (a) $2x - y + 3z = 9$ and $\frac{1}{2}x + y - z = 0$.
- ☐ (b) $4x - y + 3z = 9$ and $4x - y + 3z = 0$.
- ☐ (c) $x + y - 10z = 1$ and $5x + 5y + z = 11$.
- ☐ (d) $3x - 3y + 3z = 8$ and $x - y + z = 8$.

2.7 Problem A7

Choose the correct statement:

- ☐ (a) Every field is a ring.
- ☐ (b) Cardinality of $\mathbb{Q}$ is greater than cardinality of $\mathbb{N}$.
- ☐ (c) Cardinality of $\mathbb{Q}_+$ (nonnegative rationals) is smaller than cardinality of $\mathbb{Z}$.
- ☐ (d) The inner product of two vectors in $\mathbb{R}^3$ cannot be greater than 1.

2.8 Problem A8

Let $z = \frac{\sqrt{3}}{2} - \frac{1}{2}i$. Find z^6.

- ☐ (a) -1
- ☐ (b) 0
- ☐ (c) 1
- ☐ (d) i
- ☐ (e) $-i$

2.9 Problem A9

Consider the set $S = \{\frac{n+1}{n} : n \in \mathbb{N}\}$ in $\mathbb{R}$ (with the usual order). Find $\inf S + \sup S$.

- ☐ (a) 0
- ☐ (b) 2
- ☐ (c) 3
- ☐ (d) $+\infty$

2.10 Problem A10

The English alphabet contains 21 consonants and 5 vowels. How many strings of six lowercase letters of the English alphabet contain exactly one vowel?

- ☐ (a) $5 \cdot (21C5)^5$
- ☐ (b) $5 \cdot 21^5$
- ☐ (c) $30 \cdot 21^5$
- ☐ (d) $5 \cdot 21 \cdot 20 \cdot 19 \cdot 18 \cdot 17$

3 Part B: MULTIPLE CHOICE QUESTIONS

- **Instructions**: Each question is worth 2 points. There may be multiple correct choices. To receive points, you must select every correct option and avoid selecting any incorrect ones. No partial credit is given.

3.1 Problem B1

Which relations below are equivalence relations?

- ☐ (a) R is a relation on $\mathbb{Z}$ such that xRy if and only if $x^2 \equiv y^2 \pmod 5$.
- ☐ (b) R is a relation on $\mathbb{Z}$ such that xRy if and only if $2 \mid x^2 + 2y^2$.
- ☐ (c) R is the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $ad = bc$.

3.2 Problem B2

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = 4 - x^2$. Find $f^{-1}(B)$ if $B = (2, 4)$. Answer:

3.3 Problem B3

Choose all correct statements below (numbers a, b, c, n are natural)

- ☐ (a) If 3 divides $n > 2026$, then $3^2 \mid n$.
- ☐ (b) If $5^2 \mid ab$, then $5 \mid (a + b)$.
- ☐ (c) If $a \mid b^2$ and $b \mid c^3$, then $a \mid c^6$.
- ☐ (d) If $15 \mid n$ and $35 \mid n$, then $(15 \cdot 35) \mid n$.
- ☐ (e) If $6 \mid n$ and $35 \mid n$, then $(6 \cdot 35) \mid n$.

3.4 Problem B4

Choose which functions below are injective

- ☐ (a) $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2 - 2x + 4$.
- ☐ (b) $f : [-1, 1] \to \mathbb{R}, f(x) = x^3$.
- ☐ (c) $f : (0, 1) \to \mathbb{R}_+, f(x) = \frac{1}{x}$.
- ☐ (d) $f : [2, +\infty] \to \mathbb{R}, f(x) = \sqrt{x^2 - 3x + 2}$.

3.5 Problem B5

Real sequences (a_n), (b_n), and (c_n) are such that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$ and $\lim_{n\to\infty} c_n = 1$. What is $\lim_{n\to\infty} \frac{c_n}{a_n - b_n}$?

- ☐ (a) $-\infty$.
- ☐ (b) $+\infty$.
- ☐ (c) Either $-\infty$ or $+\infty$, the sign is undetermined.
- ☐ (d) The limit is undetermined.
- ☐ (e) 0

3.6 Problem B6

Write $0.2\overline{1} + 0.\overline{9}$ as a fraction. Answer:

3.7 Problem B7

Find the remainder from dividing 2026^{2026} by 7. Answer:

3.8 Problem B8

The following code aims to compute the Cartesian product C of two lists A and B. Fill in the missing bits.

```
function productLists(A, B)
    C = []
    for a in A
        for b in B
            ?
        end
    end
    ?
end
```

3.9 Problem B9

What is 69 in a positional system with base 7? Answer:

3.10 Problem B10

Given a list a, we want to make a list b that is reversed to a. Complete the missing parts to achieve this.

```
function makeReverse(a)
    n = length(a)
    b = zeros(n)
    for

    end
    return b
end
```

4 Part C: OPEN QUESTIONS

- **Instructions**: Each problem is worth 3 points. Partial credit may be given. Detailed solutions are required for all problems.

4.1 Problem C1

A group of seven people has a combined age of 332 years. Prove that it is possible to select three people from this group whose combined age is at least 142 years.

4.2 Problem C2

Let $P(x) = (x^2 - x + 1) \cdot (x^3 + 2x^2 - 1)^2 \cdot (x^4 + 3x^2 - 3)^3$. Find the sum of its coefficients.

4.3 Problem C3

Prove that for all $n > 1$, $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$.

4.4 Problem C4

Find the limit $\lim_{n\to\infty} \left(\frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} \right)$.